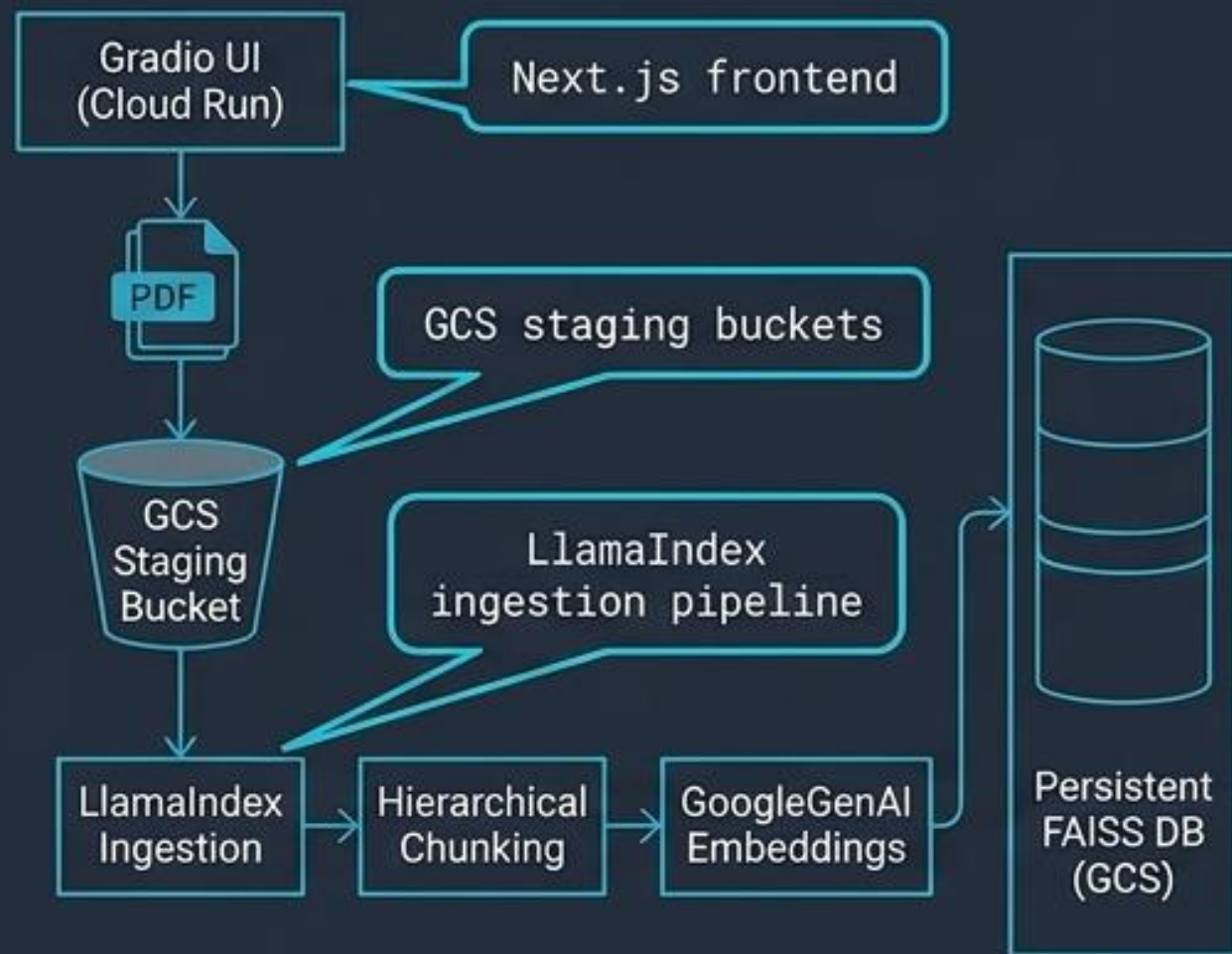


Enterprise Cloud-Native System Architecture

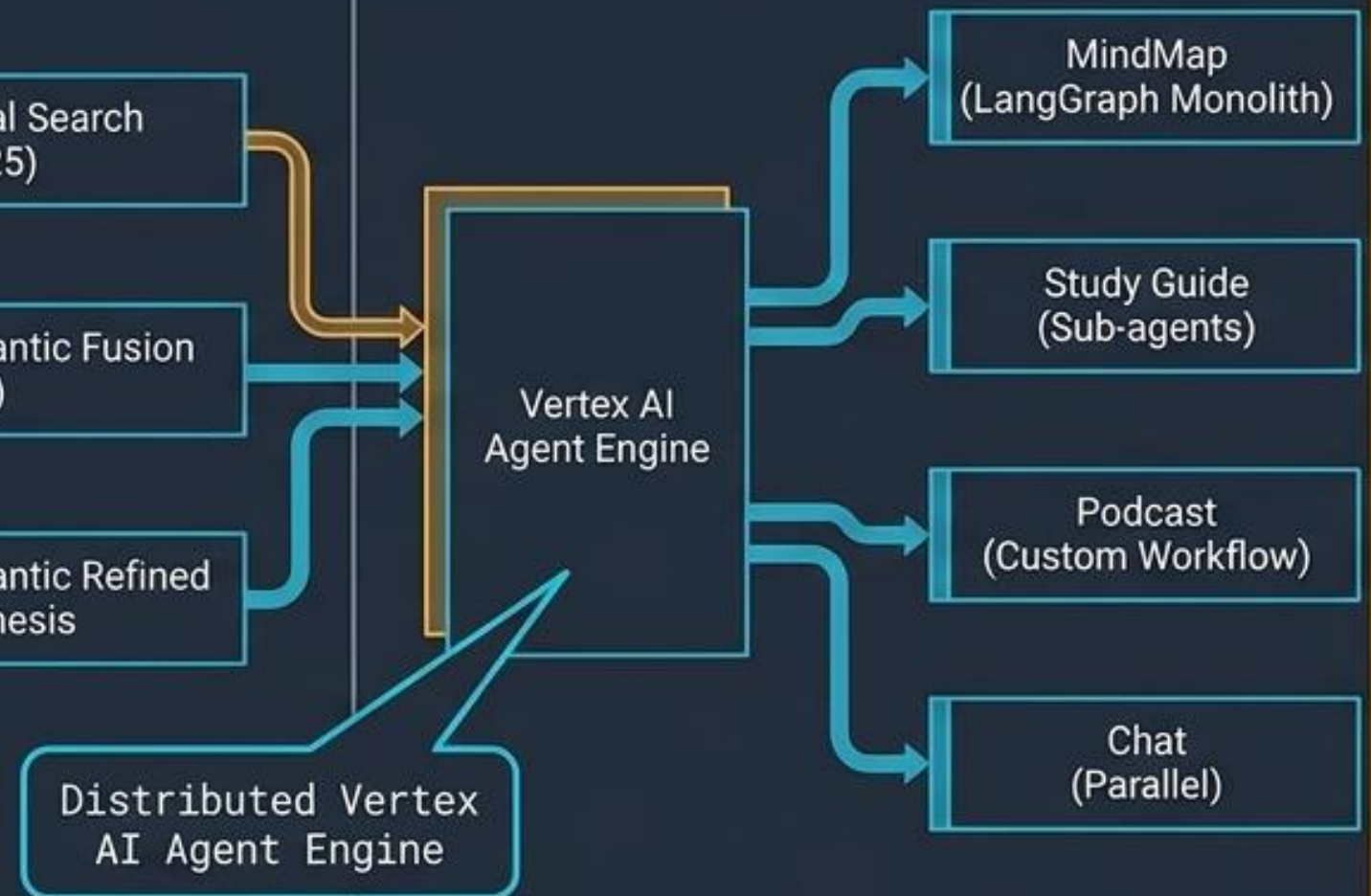
Ingestion & State



Retrieval (The 3 Paths)

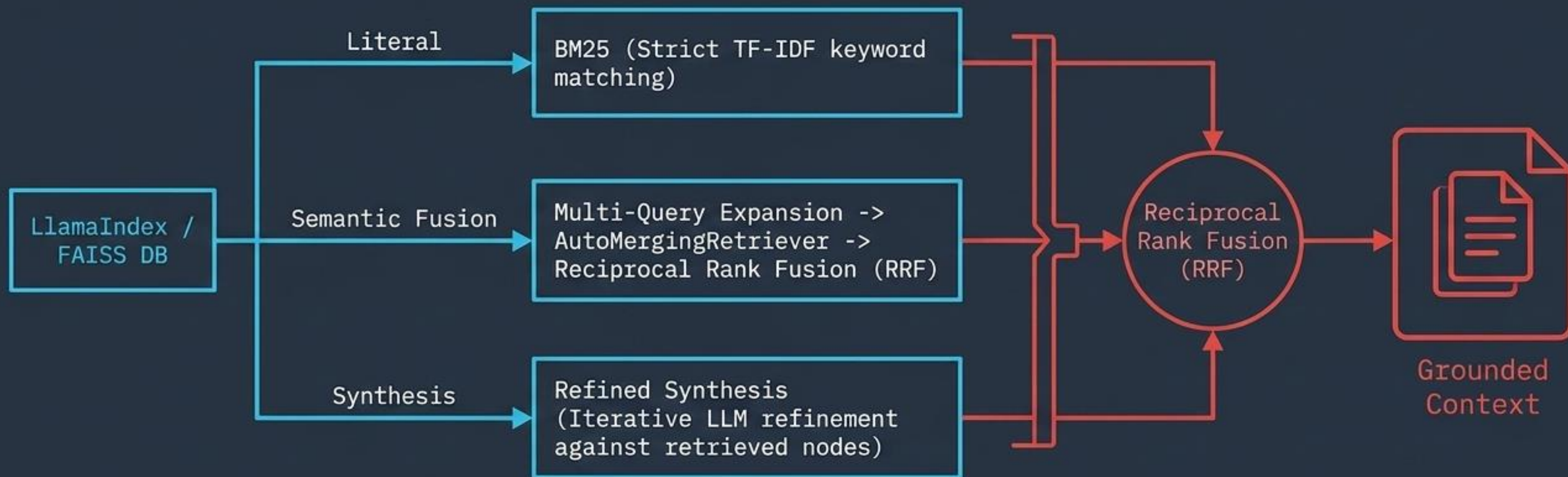


Intelligence (Vertex AI)



The Initial State: Conceptually pure during local validation, but destined to clash with serverless production physics.

Eliminating Hallucinations via Tri-Path Search



Dynamic Parallel Querying

Fault-Tolerant Hallucination Elimination

Tech Stack

LlamaIndex

GoogleGenAI Embeddings

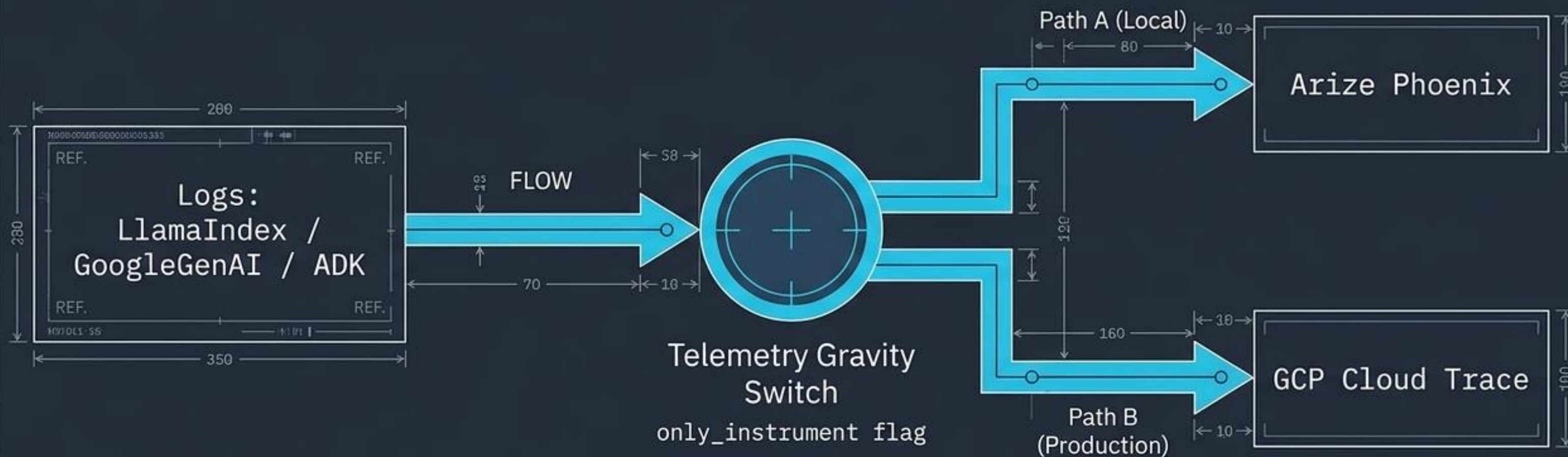
FAISS

BM25

RRF

Telemetry Gravity and the Unified Observability Pipeline

A single flag dynamically hijacks the instrumentation pipeline, routing granular spans to local or production environments without altering the execution code.



Tech Stack

OpenTelemetry

Arize Phoenix

GCP Cloud Trace

Python Iterators

High-Fidelity Custom Quality Assurance Evaluations



Standard ADK Registry Limitation

Black-Box Failure: Built-in evaluators completely crashed when introspecting custom LangGraph workflows.

Replaced 287 lines of broken framework code with a 10-line Python iterator. Achieved 91% aggregate pass rate.



Custom JSON Rubric Evaluator

Engineered an "LLM-as-a-Judge" pipeline against strict manual JSON rubrics.

Tech Stack

LLM-as-a-Judge

ADK Evaluators

Custom JSON Rubrics

Gemini 2.5 Pro

FinOps Governance and the Nuclear Teardown Protocol



Vertex AI Agent Engine continuously billed for "warm" Zombie reasoning containers during downtime.

The Protocol

- ✓ 1. Scale Cloud Run to zero.
- ✓ 2. Delete Vertex AI reasoning engines via SDK.
- ✓ 3. Purge Artifact Registry builds.
- ✓ 4. Revoke IAM and API access permissions.

Tech Stack

Python SDK

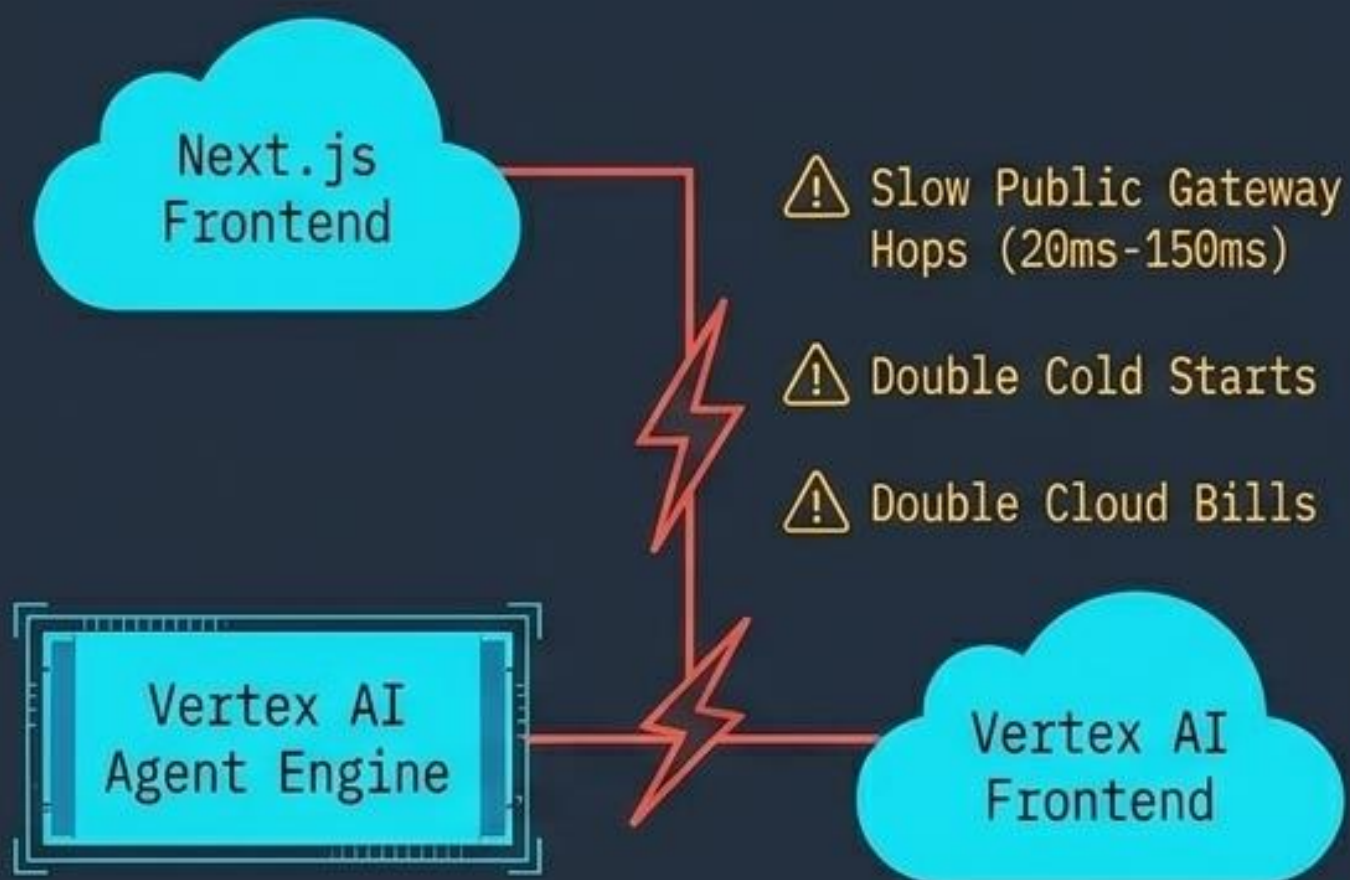
GCP Billing

FinOps Automation

IAM Restructuring

The Enterprise Pivot: Single-Container Monolith Topology

Before (Scattered Islands)



After (Unified Fortress)



Tech Stack

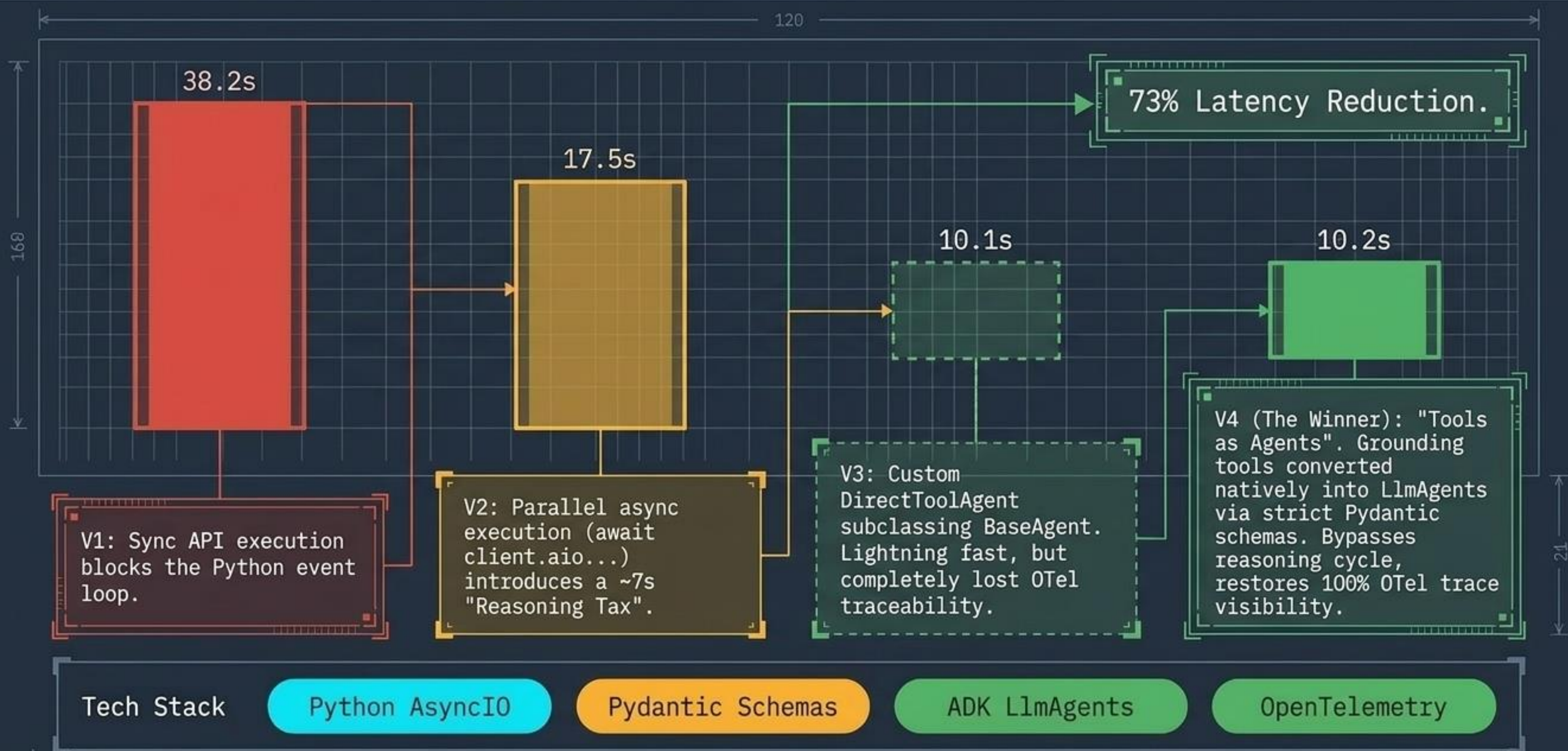
Multi-Stage Docker

POSIX Process Multiplexer

Private TCP Loopback

Unix Init Traps

The RAG Latency Optimization Waterfall



Hybrid Networking: The Cloud-to-Edge Bridge

3 Min Generation for 20 Min Audio



Symptom:

Long-Running Operations (LROs) causing massive 504 Deadline Exceeded timeouts.

Remedy:

Ejected heavy MLX-Audio synthesis to isolated FastMCP server via self-polling architecture. Tuned polling intervals (2s normal / 8s error) to prevent proxy DDOS crashes.

Tech Stack

FastMCP

ZioK Reserved Tunnel

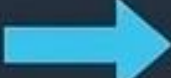
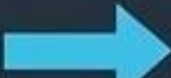
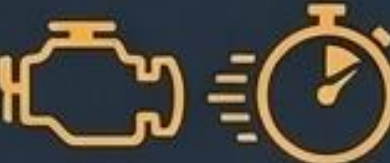
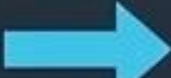
Server-Sent Events (SSE)

MLX-Audio

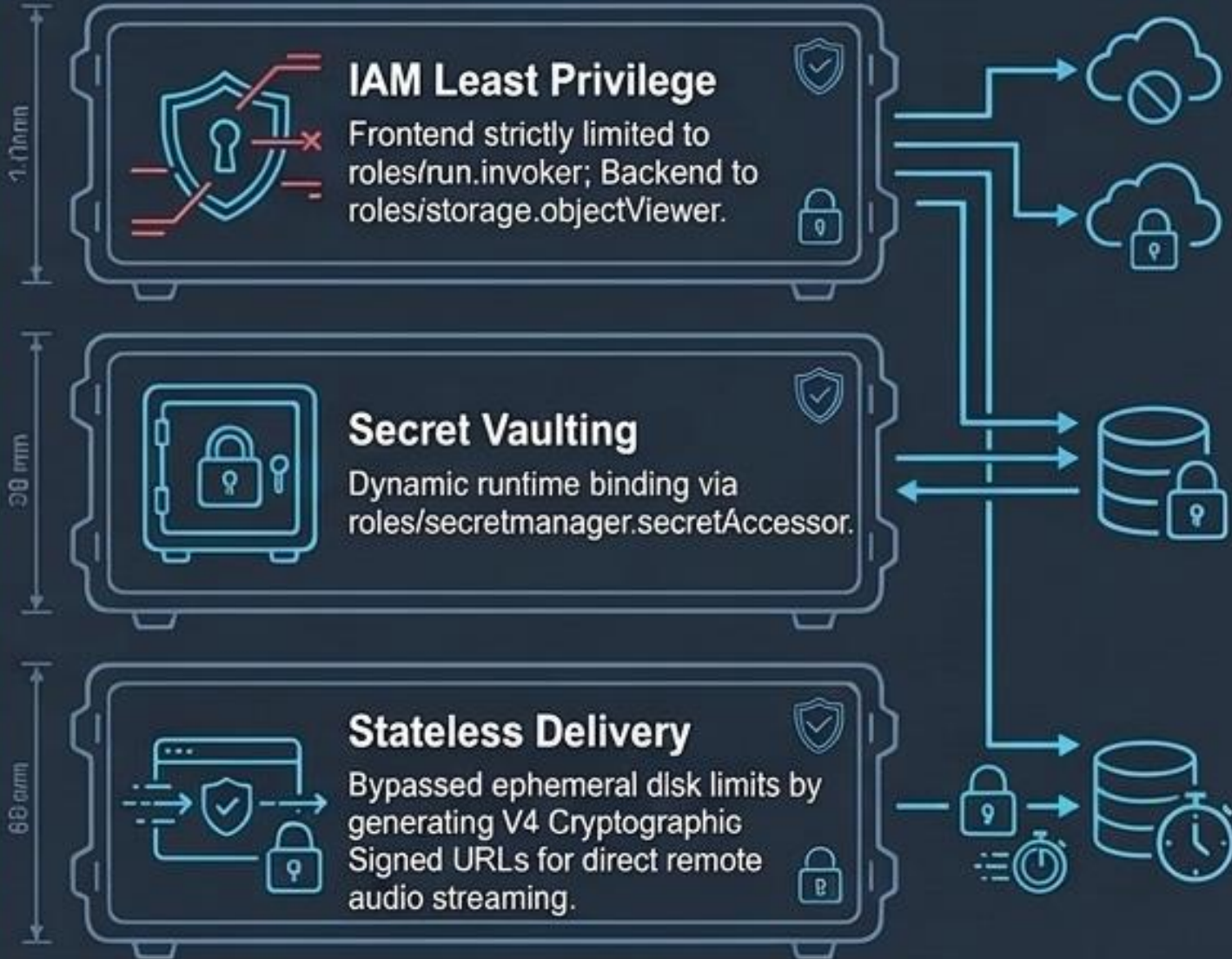
Apple Silicon

Conquering Container Physics and Zero-Trust Security

Physics Diagnostic Matrix

CHALLENGE:  Port Collisions		SOLUTION:  UNIFIED, ALIGNED NETWORK FLOW Fixed via Ingress Realignment (forcing frontend-to-backend API calls over a sub-millisecond private loopback).
CHALLENGE:  Unexpected Process Termination		SOLUTION:  SIGNAL SHIELDING Fixed via Signal Shielding (wrapper preventing silent background process death).
CHALLENGE:  Slow Service Startup		SOLUTION:  10s PREDICTABLE STARTUP SYNC Fixed via a 10s Predictable Startup Sync within the container lifecycle.

Security Architecture Vaults



Tech Stack

Linux POSIX Signals

Port Mapping Alignment

Boot Synchronization

Zero-Trust Local Routing